

GAUTHAM PRAVEEN

gauthampraveen76@gmail.com | +91 9400722621 | GitHub | LinkedIn

EDUCATION

VIT Vellore

B.Tech in Computer Science with specialization in AI/ML

– Grade: 9.42

Vellore, Tamil Nadu

2023 – 2027

Devagiri CMI Public School

Higher Secondary – Grade: 96.4

Secondary – Grade: 98.2

Kozhikode, Kerala

2023

2021

EXPERIENCE

CodeChef - VIT Student Chapter

Senior Core - Developer

- Developed responsive frontend interfaces and a feature-rich Chrome extension to enhance the functionality of our college website
- Contributed to hardware prototyping using Arduino and co-organized Clueminati'24 during Gravitas, the annual tech fest

Vellore, Tamil Nadu

March 2024 – Present

PROJECTS

Basketball Analytics – Computer Vision

March 2026

- Built an end-to-end CV pipeline processing basketball footage using YOLOv5 and ByteTrack for player and ball tracking across frames.
- Integrated FashionCLIP for jersey-based team classification and RF-DETR via Roboflow to detect shot outcomes in real time.
- Automated game analytics including ball possession, passes, interceptions, jersey number OCR, and player speed metrics overlaid on annotated output video.

RAG query extractor

June 2025

- Built a Streamlit-based intelligent database Q&A app that integrates Google's Gemini model with a private SQL database using LangChain.
- Used few-shot prompting and FAISS-based semantic retrieval to guide the LLM in generating accurate, context-aware SQL queries from natural language questions.
- Executed generated queries via SQLAlchemy and returned real-time answers through an interactive Streamlit interface.

Public Health Clinic Web Application

April 2025

- Register/login, manage profile, book appointments with available doctors, and modify appointment details
- Integrated a fine-tuned DistilBERT classifier trained on synthetic health data to interpret user symptom queries and predict possible diseases from a limited common diseases set
- Emphasized clean UI/UX, responsive design, and seamless backend integration with MongoDB for data persistence

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, HTML, CSS, JavaScript, R

AI/ML: Exploratory Data Analysis, Scikit-Learn, Pytorch, Transformers, FineTuning LLMs, Langchain, RAG, LangGraph, MCPs

Web: React JS, Node JS, Fast API, ExpressJS, Streamlit, REST APIs

Tools: Git, GitHub, VS Code, Jupyter Notebook, Conda, Postman

Databases: SQL, MongoDB, Supabase, FAISS, ChromaDB

Other: Data Structures and Algorithms, Arduino IDE, Hardware Technologies, n8n

CERTIFICATIONS

- **GEN AI Using IBM WatsonX** – IBM
- **Fundamentals of Deep Learning** – NVIDIA
- **Machine Learning Specialization** – DeepLearning.AI, Stanford University

Issued Jun 2025

Issued Sep 2024

Issued Jul 2024

ACHIEVEMENTS

- **1st Place – ACM Code2Create Hackathon 2025**, VIT Vellore (48-hour Hackathon). Developed **Awaaz**, a VR + AI presentation rehearsal tool with real-time NPC questioning and personalized post-presentation feedback. Implemented end-to-end workflow: presentation upload → VR simulation → speech analysis → feedback report. *Tech stack: React.js, Node, Express, Unity (OpenXR), Supabase, OpenAI API, Eleven Labs TTS.*
- Registration and Reception Volunteer at **Gravitas'24** annual tech fest at VIT
- Participated in **RoboFest 2018 at Lawrence Tech University**, Michigan, USA, as part of a team competing in Bottle Sumo event
- Winner of district-level **IoT exhibition** in 2019, developing a Smart Agricultural Irrigation System using NodeMCU with soil moisture and humidity sensors. The system provided real-time crop health insights and enabled farmers to automate irrigation